



rates of reaction

Name: _____

Class: _____

Date: _____

Time:

Marks:

65 marks

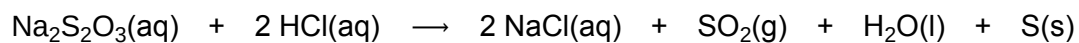
Comments:

1.

Sodium thiosulfate solution reacts with dilute hydrochloric acid.

The solution becomes cloudy as the reaction takes place.

(a) The equation for the reaction is:



Explain why the solution becomes cloudy.

(2)

- Your plan should give valid results.

[illegible]

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2.

This question is about ammonia and fertilisers.

Ammonia is produced from nitrogen and hydrogen.

A catalyst is used to speed up the reaction.

The word equation for the reaction is:



(a) What does the symbol $\rightleftharpoons$ show about the reaction?

(1)

(b) Which catalyst is used when ammonia is produced from nitrogen and hydrogen?

Tick (✓) **one** box.

Chlorine

☐

Iron

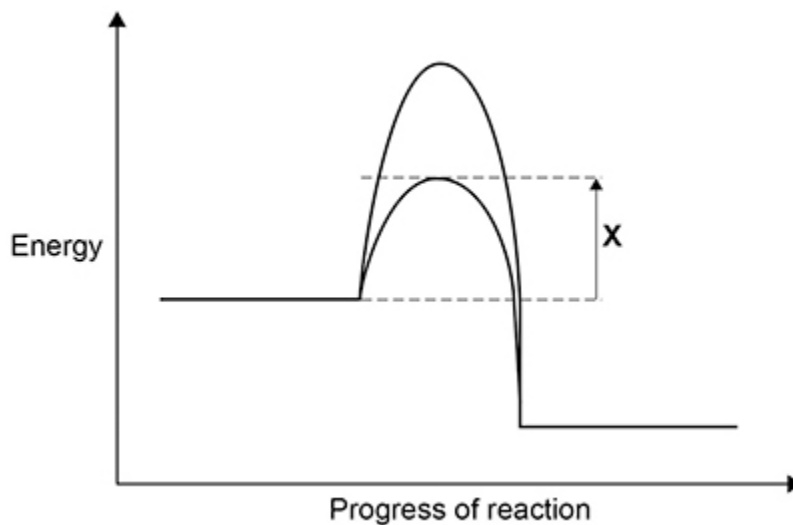
☐

Oxygen

☐

(1)

- (c) The diagram below shows the reaction profile for the production of ammonia both with a catalyst and without a catalyst.



What is represented by label X?

Tick (✓) **one** box.

Activation energy with a catalyst

☐

Activation energy without a catalyst

☐

Overall energy change with a catalyst

☐

Overall energy change without a catalyst

☐

(1)

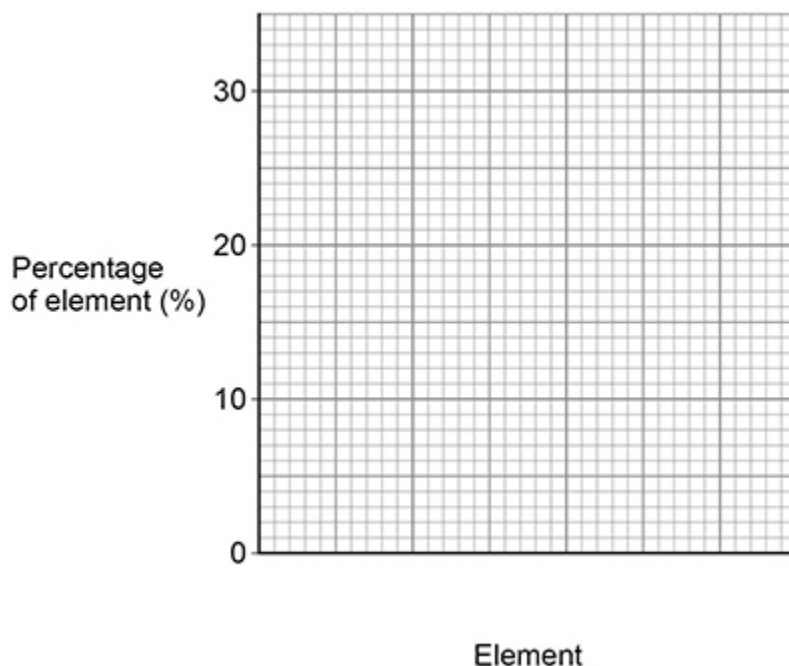
Ammonia is used to produce fertilisers.

NPK fertilisers contain the elements nitrogen, phosphorus and potassium.

A fertiliser contains:

- 22% phosphorus
- 25% potassium.

- (d) Draw a bar chart on the graph below to show the percentages of phosphorus and of potassium in this fertiliser.



(2)

- (e) Why do the percentages of phosphorus and of potassium in this fertiliser **not** add up to 100%?

(1)

Fertilisers help plants grow by adding essential elements to soil.

The table below shows the percentages of nitrogen, phosphorus and potassium in four fertilisers, **A, B, C and D**.

Fertiliser	Percentage (%) of essential element		
	Nitrogen (N)	Phosphorus (P)	Potassium (K)
A	14	0	39
B	25	16	23
C	21	23	0
D	21	0	0

(f) Plants lacking essential elements do not grow well because:

- too little phosphorus can cause slow plant growth
- too little potassium can cause leaves to have brown edges.

Which fertiliser helps prevent slow plant growth **and** brown leaf edges?

Use the table above.

Tick (✓) **one** box.

A	☐	B	☐	C	☐	D	☐
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(1)

(g) Which fertiliser has the greatest total percentage of essential elements?

Use the table above.

Tick (✓) **one** box.

A	☐	B	☐	C	☐	D	☐
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(1)

(Total 8 marks)

3.

This question is about the rate of the reaction between hydrochloric acid and calcium carbonate.

A student investigated the effect of changing the size of calcium carbonate lumps on the rate of this reaction.

This is the method used.

1. Pour 40 cm³ of hydrochloric acid into a conical flask.
2. Add 10.0 g of small calcium carbonate lumps to the conical flask.
3. Attach a gas syringe to the conical flask.
4. Measure the volume of gas produced every 30 seconds for 180 seconds.
5. Repeat steps 1 to 4 using 10.0 g of large calcium carbonate lumps.

The student calculated the number of moles of gas from each volume of gas measured.

The table below shows the student's results for large calcium carbonate lumps.

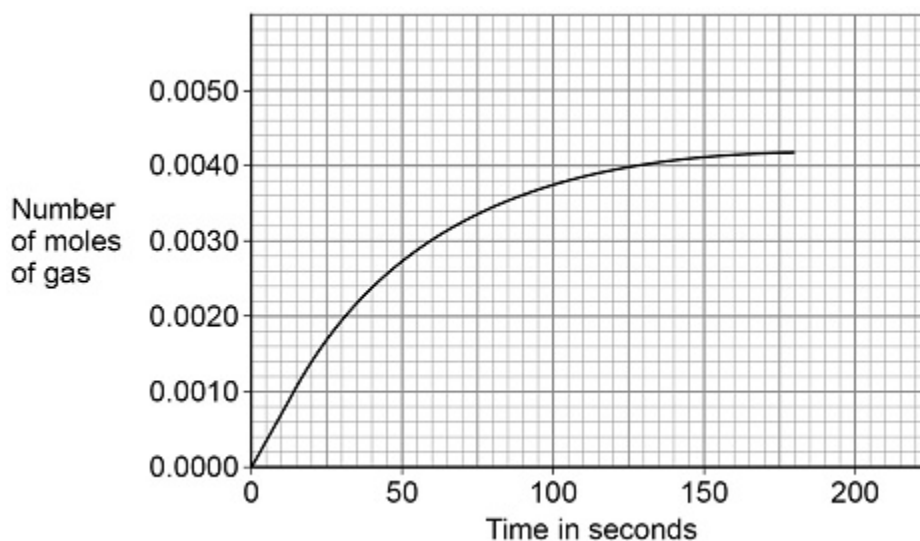
Time in seconds	Number of moles of gas
0	0.0000
30	0.0011
60	0.0020
90	0.0028
120	0.0034
150	0.0038
180	0.0040

The student plotted the results for small calcium carbonate lumps on the graph below.

(a) Complete the graph below.

You should:

- plot the data for large calcium carbonate lumps from the table above
- draw a line of best fit.



(3)

(b) Determine the mean rate of reaction for **small** calcium carbonate lumps between 20 seconds and 105 seconds.

Give the unit.

Use the graph above.

Mean rate of reaction = _____ Unit _____

(4)

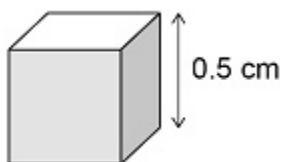
- (c) The student concluded that the large calcium carbonate lumps reacted more slowly than the small calcium carbonate lumps.

How do the student's results show that this conclusion is correct?

(1)

The difference in the rates of reaction of large lumps and of small lumps of calcium carbonate depends on the surface area to volume ratios of the lumps.

The diagram below shows a cube of calcium carbonate.



- (d) Calculate the surface area to volume ratio of the cube in above diagram.

Give your answer as the simplest whole number ratio.

Surface area : volume = _____ : _____

(3)

- (e) A larger cube of calcium carbonate has sides of 5 cm

Describe how the surface area to volume ratio of this larger cube differs from that of the cube shown in the diagram above.

(1)

(Total 12 marks)

4.

This question is about displacement reactions.

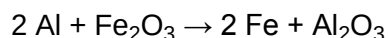
- (a) The displacement reaction between aluminium and iron oxide has a high activation energy.

What is meant by 'activation energy'?

(1)

- (b) A mixture contains 1.00 kg of aluminium and 3.00 kg of iron oxide.

The equation for the reaction is:



Show that aluminium is the limiting reactant.

Relative atomic masses (A_r): O = 16 Al = 27 Fe = 56

(4)

Magnesium displaces zinc from zinc sulfate solution.

- (c) Complete the ionic equation for the reaction.

You should include state symbols.



(2)

- (d) Explain why the reaction between magnesium atoms and zinc ions is both oxidation and reduction.

(2)

(Total 9 marks)

5.

During a thunderstorm lightning strikes the Eiffel Tower.



By M. G. Loppé [Public domain], via Wikimedia Commons

In lightning the temperature can reach 30 000 °C. This causes nitrogen and oxygen in the air to react, producing nitrogen oxide. This reaction has a high *activation energy* and is *endothermic*.

- (a) Nitrogen and oxygen in the air do not react easily.

What makes nitrogen and oxygen react during thunderstorms?

(1)

- (b) Complete the word equation for the reaction of nitrogen with oxygen.

nitrogen + _____ → _____

(1)

- (c) In an *endothermic* reaction, energy is taken in from the surroundings.

Draw a ring around the correct answer to complete the sentence.

In an *endothermic* reaction, the energy needed to break existing bonds is

less than

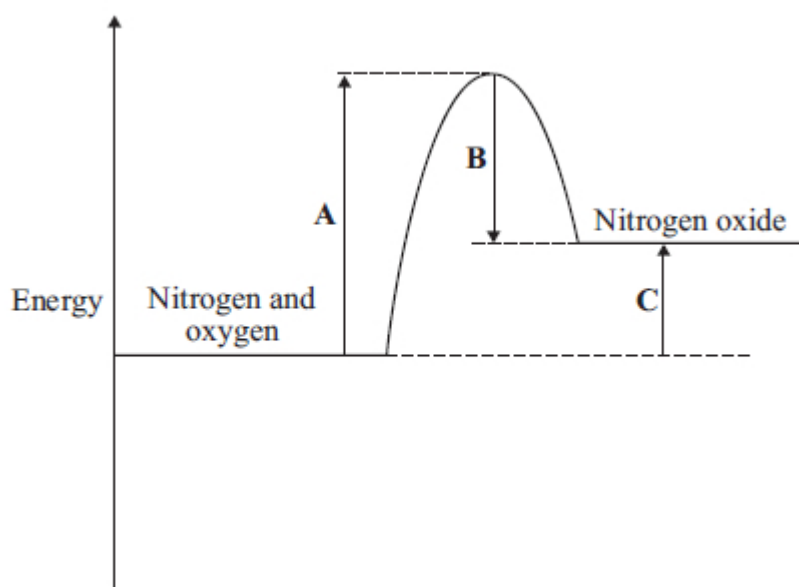
more than

the same as

the energy released from forming new bonds.

(1)

- (d) The energy level diagram for this reaction is shown.



Use the energy level diagram to help you to answer these questions.

- (i) Which energy change, **A**, **B** or **C**, represents the *activation energy*?

(1)

(ii) Which energy change, **A**, **B** or **C**, shows that this reaction is *endothermic*?

(1)

(Total 5 marks)

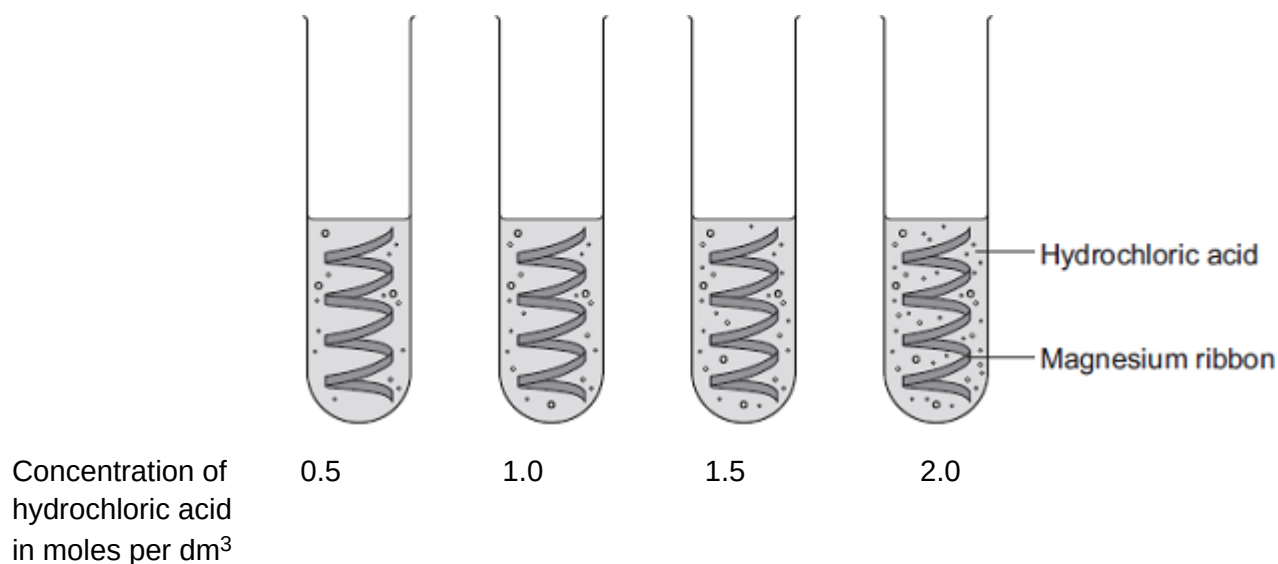
6.

A student investigated the rate of reaction of magnesium and hydrochloric acid.



The student studied the effect of changing the concentration of the hydrochloric acid.

She measured the time for the magnesium to stop reacting.



(a) The student changed the concentration of the hydrochloric acid.

Give **two** variables that the student should control.

1. _____

2. _____

(2)

(b) (i) The rate of reaction increased as the concentration of hydrochloric acid increased.

Explain why.

(2)

- (ii) Explain why increasing the temperature would increase the rate of reaction.

(3)

- (c) (i) The student had a solution of sodium hydroxide with a concentration of 0.100 moles per dm³.

She wanted to check the concentration of a solution of hydrochloric acid.

She used a pipette to transfer 5.00 cm³ of the hydrochloric acid into a conical flask.

She filled a burette with the 0.100 moles per dm³ sodium hydroxide solution.

Describe how she should use titration to obtain accurate results.

(4)

- (ii) Sodium hydroxide neutralises hydrochloric acid as shown in the equation:



The student found that 27.20 cm³ of 0.100 moles per dm³ sodium hydroxide neutralised 5.00 cm³ of hydrochloric acid.

Calculate the concentration of the hydrochloric acid in moles per dm³.

Give your answer to three significant figures.

Concentration of hydrochloric acid = _____ moles per dm³

(3)

(Total 14 marks)

7.

A student investigated how temperature affects the rate of reaction between magnesium carbonate and dilute hydrochloric acid.

This is the method used.

1. Heat hydrochloric acid to 30 °C in a conical flask.
2. Add magnesium carbonate powder to the conical flask.
3. Measure the loss in mass of the flask and contents every 20 seconds for 140 seconds.
4. Repeat steps 1-3 with hydrochloric acid heated to 50 °C

- (a) Explain why the contents of the conical flask lose mass.

(2)

- (b) The table below shows the student's results for hydrochloric acid at 30 °C

Time in seconds	Loss of mass in grams
0	0.00
20	0.26
40	0.48
60	0.67
80	0.82
100	0.91
120	0.96
140	0.99

Plot the data from the table above on **Figure 1**.

Draw a line of best fit.

Figure 1

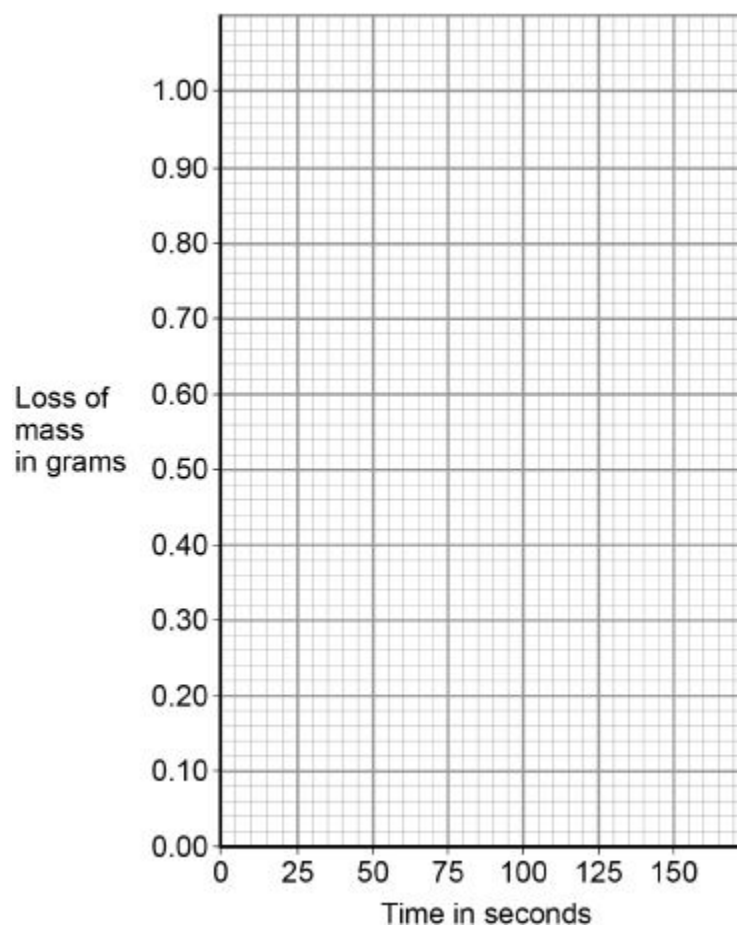
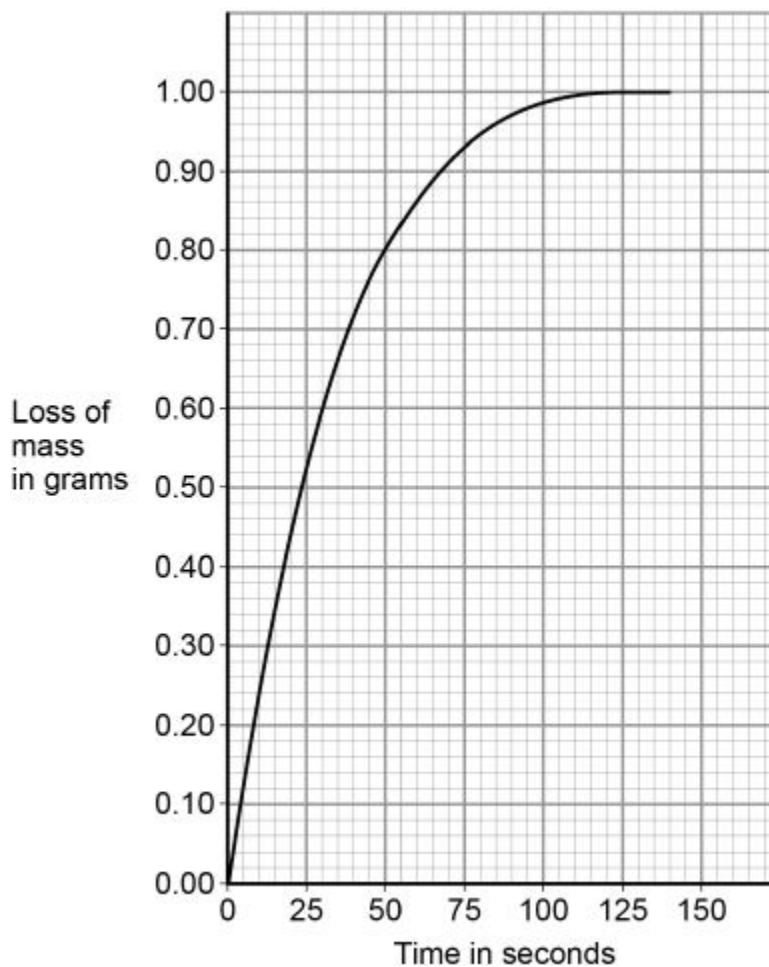


Figure 2 shows the student's results for hydrochloric acid at 50 °C

Figure 2



- (c) Determine the rate of reaction at 50 °C when the loss of mass is 0.95 g

Show your working on **Figure 2**.

Give your answer to 2 significant figures.

Rate of reaction = _____ g/s

(4)
(Total 9 marks)

Mark schemes

1.

(a) sulfur (formed)

allow S / S₈ (formed)

1

(which is a) precipitate

allow (which is a) solid

allow (which is) insoluble

1

(b) **Level 3:** The method would lead to the production of a valid outcome. All key steps are identified and logically sequenced.

5–6

Level 2: The method would not necessarily lead to a valid outcome. Most steps are identified, but the method is not fully logically sequenced.

3–4

Level 1: The method would not lead to a valid outcome. Some relevant steps are identified, but links are not made clear.

1–2

No relevant content

0

Indicative content

method

- measure (indicated) volume of sodium thiosulfate
- place sodium thiosulfate in (conical) flask
- measure (indicated) volume of hydrochloric acid
- place on cross or between light sensor
- or
- connect to a gas syringe
- or
- other suitable method for timing a change
- add hydrochloric acid to (conical) flask
- swirl
- start stopclock / stopwatch
- measure time for cross to become no longer visible
- or
- log light transmission over time
- or
- measure time for fixed volume of gas to be produced
- repeat and find mean
- repeat for different concentrations of sodium thiosulfate
- or change ratio of sodium thiosulfate volume : water volume

control variables

- concentration of hydrochloric acid
- volume of hydrochloric acid
- (total) volume of sodium thiosulfate solution

- 2.** (a) (the reaction is) reversible
allow description of a reversible reaction 1
- (b) iron 1
- (c) activation energy with a catalyst 1
- (d) bar to 22 (%) labelled phosphorus / P
allow a tolerance of $\pm \frac{1}{2}$ a small square 1
- bar to 25 (%) labelled potassium / K
if no other mark is awarded, allow 1 mark for two bars drawn to 22% and 25% 1
- (e) there are other elements in the fertiliser (besides phosphorus and potassium)
or
there is nitrogen in the fertiliser
allow there are other substances in the fertiliser (besides phosphorus and potassium) 1
- (f) **B** 1
- (g) **B** 1
- [8]
- 3.** (a) all seven points plotted correctly
allow a tolerance of $\pm \frac{1}{2}$ small square
allow 1 mark for five or six points plotted correctly 2
- line of best fit 1

(b) 0.0038 **and** 0.0014

1

$$\frac{0.0038 - 0.0014}{105 - 20}$$

allow correct use of incorrectly determined mole value(s)

1

$$= 0.000028$$

or

$$= 2.8 \times 10^{-5}$$

1

mol/s

allow moles per second

1

- (c) (for large lumps) a smaller number of moles of gas is collected in the same time
or
(for large lumps) more time is needed to collect the same number of moles of gas
or
the line (of best fit for large lumps) is less steep

allow converse statement for small lumps

allow the line (of best fit for large lumps) takes more time to become horizontal

1

- (d) (surface area = $6 \times 0.5 \times 0.5$) = $1.5 \text{ (cm}^2\text{)}$

1

$$(\text{volume} = 0.5 \times 0.5 \times 0.5) = 0.125 \text{ (cm}^3\text{)}$$

1

$$(\text{surface area} : \text{volume} =) 12 : 1$$

allow correctly calculated ratio using incorrectly calculated values for surface area and/or volume

1

- (e) decreases by a factor of 10

allow 10 times smaller

allow one tenth

allow 1/10

allow 1 : 10 (large cube to small cube)

1

[12]

4.

- (a) the (minimum) energy needed for particles to react
or

the (minimum) energy needed for a reaction to occur

allow the (minimum) energy needed to start a reaction

1

(b) (M_r of Fe_2O_3 =) 160

1

$$(\text{moles } \text{Fe}_2\text{O}_3 = \frac{3000}{160} =)$$

18.75 (mol)

allow correct use of incorrectly calculated M_r

1

$$(\text{moles Al} = \frac{1000}{27} =) 37.0 \text{ (mol)}$$

allow 37.037037 (mol) correctly rounded to at least 2 significant figures

*if both MP2 and MP3 are not awarded allow 1 mark for 0.01875 mol Fe_2O_3 **and** 0.037 mol Al*

1

(aluminium is limiting because)

37.0 mol is less than the ($2 \times 18.75 =$) 37.5 mol (aluminium needed)

or

iron oxide is in excess because 18.75 mol is more than the ($\frac{37.0}{2} =$) 18.5 mol (iron oxide needed)

allow correct use of incorrect number of moles from steps 2 and/or 3

alternative approaches:

approach 1:

(finding required mass of aluminium by moles method)

(M_r of Fe_2O_3 =) 160 (1)

$$(\text{moles } \text{Fe}_2\text{O}_3 = \frac{3000}{160} =)$$

18.75 (mol) (1)

allow correct use of incorrectly calculated M_r

(moles Al needed = $18.75 \times 2 =$) 37.5 (mol)

and

(mass Al needed = $37.5 \times 27 =$) 1012.5 (g) **or** 1.0125 kg (1)

allow correct use of incorrectly calculated moles of iron oxide

allow correct use of incorrectly calculated moles of aluminium needed

(so) 1.00 kg of aluminium is not enough (1)

dependent on calculated mass of aluminium needed being greater than 1.00 (kg)

approach 2:**(finding required mass of aluminium by proportion method)** $(M_r \text{ of } \text{Fe}_2\text{O}_3 \Rightarrow) 160 \text{ (1)}$ $(3.00 \text{ kg } \text{Fe}_2\text{O}_3 \text{ needs})$ $\frac{3.00}{160} \times 2 \times 27 \text{ (kg Al) (1)}$ *allow correct use of incorrectly calculated M_r* $(\Rightarrow) 1.0125 \text{ (kg) (1)}$ $(\text{so}) 1.00 \text{ kg of aluminium is not enough (1)}$ *dependent on calculated mass of aluminium needed
being greater than 1.00 (kg)***alternative approaches:****approach 3:****(finding required mass of iron oxide by moles method)** $M_r \text{ of } \text{Fe}_2\text{O}_3 \Rightarrow 160 \text{ (1)}$ $(\text{moles Al} = \frac{1000}{27} \Rightarrow) 37.0 \text{ (mol) (1)}$ *allow 37.037037 (mol) correctly rounded to at least 2 significant
figures* $(\text{moles } \text{Fe}_2\text{O}_3 \text{ needed}) = \frac{37.0}{2} = 18.5 \text{ (mol)}$ **and** $(\text{mass } \text{Fe}_2\text{O}_3 \text{ needed} = 18.5 \times 160 \Rightarrow) 2960 \text{ (g) or } 2.96 \text{ (kg) (1)}$ *allow correct use of incorrectly calculated moles of
aluminium**allow correct use of incorrectly calculated moles of iron
oxide needed**allow correct use of incorrectly calculated M_r* $(\text{so}) 3.00 \text{ kg of iron oxide is an excess (1)}$ *dependent on calculated mass of iron oxide needed
being less than 3.00 (kg)*

approach 4:
(finding required mass of iron oxide by proportion method)

(M_r of Fe_2O_3 =) 160 (1)

(1.00 kg Al needs) $\frac{1.00}{2 \times 27}$ (kg Fe_2O_3) (1)

allow correct use of incorrectly calculated M_r

(=) 2.96 (kg) (1)

(so) 3.00 kg of iron oxide is an excess (1)

*dependent on calculated mass of iron oxide needed
being less than 3.00 (kg)*

1

(c) $\text{Mg(s)} + \text{Zn}^{2+}(\text{aq}) \rightarrow \text{Mg}^{2+}(\text{aq}) + \text{Zn(s)}$

allow multiples

*allow 1 mark for $\text{Mg}^{2+} + \text{Zn}$ with missing or incorrect
state symbols*

2

(d) magnesium (atoms) are oxidised because they lose electrons

1

(and) zinc (ions) are reduced because they gain electrons

*if no other marks awarded allow 1 mark for magnesium
(atoms) lose electrons and zinc (ions) gain electrons 1*

1

[9]

5.

(a) electricity / (high) temperatures

allow lightning / heat

ignore energy

1

(b) nitrogen + oxygen $\rightarrow$ nitrogen oxide/ monoxide

allow any oxide of nitrogen

1

(c) more than

1

(d) (i) A

1

(ii) C

1

[5]

6.

(a) any **two** from:

- temperature (of the HCl)
- mass or length of the magnesium
- surface area of the magnesium
- volume of HCl

2

(b) (i) (a greater concentration has) more particles per unit volume

allow particles are closer together

1

therefore more collisions per unit time **or** more frequent collisions.

1

(ii) particles move faster

allow particles have more (kinetic) energy

1

therefore more collisions per unit time **or** more frequent collisions

1

collisions more energetic (therefore more collisions have energy greater than the activation energy) **or** more productive collisions

1

(c) (i) add (a few drops) of indicator to the acid in the conical flask

allow any named indicator

1

add NaOH (from the burette) until the indicator changes colour **or** add the NaOH dropwise

candidate does not have to state a colour change but penalise an incorrect colour change.

1

repeat the titration

1

calculate the **average** volume of NaOH **or** repeat until concordant results are obtained

1

(ii) **moles of NaOH**

$$0.10 \times 0.0272 = 0.00272 \text{ moles}$$

correct answer with or without working gains 3 marks

1

Concentration of HCl

$$0.00272 / 0.005 = 0.544$$

allow ecf from mp1 to mp2

1

correct number of significant figures

1

[14]

7.

- (a) a gas is produced

allow carbon dioxide is produced
*do **not** accept an incorrect gas*

1

(which) escapes

***max 1** mark if evaporation mentioned*

1

- (b) all eight points plotted correctly

allow a tolerance of $\pm$ half a small square.
*allow six or seven points plotted correctly for **1** mark*

2

line of best fit

1

- (c) correctly drawn tangent at 0.95 g

1

correct value for x step **and** y step from tangent

allow evidence of use of two points on tangent either on the graph or in the text

1

(rate =) $\frac{\text{value for y step}}{\text{value for x step}}$

1

correctly evaluated and rounded to 2 sig figs

allow

(rate =) $\frac{\text{value for x step}}{\text{value for y step}}$

(i.e. inverted division)

correctly evaluated and rounded to 2 sig figs

1

an incorrect answer for one step does not prevent allocation of marks for subsequent steps

[9]

Examiner reports

1.

- (a) This question was generally well done. Most students could identify that the solution became cloudy because sulfur was produced and that it was a solid. A few students however were unfamiliar with the experiment and thought that a gas was produced, and that this was the cause. 42% of students achieved both marks.
- (b) This was an 'extended response', 'level marked' question. There were three levels and the command word was 'plan'. The question was marked holistically. Most students realised that the easiest way of investigating the effect of varying the concentration of sodium thiosulfate on the rate of reaction was to carry out a 'disappearing cross' experiment, and many students could describe this well.

Some students however, just carried out the experiment at one concentration of sodium thiosulfate, or varied the concentration of the hydrochloric acid instead of or as well as that of the sodium thiosulfate. The most common cause of a student failing to achieve full marks was to omit mentioning any variables that needed to be controlled. A minority of students, having noticed that sulfur dioxide is a gaseous product, attempted to collect the sulfur dioxide or to measure the loss in mass of the flask. These approaches, though they would not work well, were given appropriate credit.

44% of students provided a level 3 response.

3.

- (a) The point that was most frequently plotted incorrectly was the final one (180, 0.0040). A number of students failed to plot (0,0). There were some examples of rulers being used to complete lines of best fit, with a number of straight lines seen. Some students produced excessively thick or 'feathered' lines. However, the majority of students were able to score at least two marks.
- (b) If either of the mole values were incorrectly determined from the graph, it was still possible to gain credit for using these values correctly. Some attempted to determine the instantaneous rate at a particular time by drawing a tangent to the curve. Units were sometimes written imprecisely e.g. moles per s, and some wrote 'm/s'. Many students were able to score at least two marks.
- (c) This part was quite well answered by many. Most students preferred to answer in terms of line steepness rather than time and moles collected.
- (d) This item was quite well answered by many students. The most common error was to fail to multiply the area of one face by six.
- (e) Only a very few students scored here, with many just responding 'smaller' rather than stating by what factor it was smaller. A few assumed that because the cube was larger, so too was the surface area : volume ratio. Only a tiny number of students gained a mark here.

4.

- (a) Most students were able to express the idea that activation energy is the minimum energy required to start a reaction. One common error was to say that it is the energy needed to make particles collide.
- (b) A variety of approaches was seen. The majority of students calculated the number of moles of both aluminium and of iron oxide and then attempted with varying degrees of success to explain why the number of moles of aluminium was insufficient for the iron oxide. Common errors were to omit to convert the mass from kilograms to grams, and/or using 54 as the ' M_r ' of aluminium. Of those who did correctly calculate the number of moles of each reactant, some had difficulty in expressing clearly the idea of 2 moles of aluminium being needed to react with 1 mole of iron oxide, and therefore 37.0 moles of aluminium is less than the 37.5 moles required, or that 18.75 moles of iron oxide is more than the 18.5 moles required. Other students started with the mass of one reagent and calculated the required mass of the other, either by the moles method or by proportion. These students found it easier to express why there was insufficient aluminium or excess iron oxide.
- (c) Few students scored both marks. Aqueous zinc was often seen as a response. Many students did not understand the idea of an ionic equation and tried to include the sulfate ion.
- (d) Only a third of students scored both marks, but a lot of students were confused about which species gained electrons and which lost them. Definitions of oxidation and reduction in terms of gain and loss of oxygen were not appropriate here and were not credited.

5.

- (a) Some candidates wrote in terms of energy or activation energy while others put down water or carbon dioxide.
- (b) While a majority of the candidates got oxygen correct, many different things appeared as the product eg lightening, sound, energy, water and nitrate oxide
- (c) Two thirds of the candidates realised that in an endothermic reaction more energy is needed to break existing bonds than is released when new bonds form.
- (d)
 - (i) Most of the candidates were able to identify 'A' as the activation energy.
 - (ii) This seemed to be a major problem for about half of the candidates. They could not correctly identify the energy change which shows the reaction is endothermic.

7.

- (a) Most students recognised that a gas is given off by the reaction. Sometimes it was incorrectly identified as hydrogen.

The 'escape' mark was less commonly awarded either because the students thought it was sufficient to state that a gas is given off or because they thought that the gas 'evaporated'.

43% of students achieved full marks.

- (b) The points on the graph were almost invariably correctly drawn within tolerance. Most found the question easy although there were a significant number who omitted the point at 0,0.

The line of best fit was problematic for a number of students. A significant number assumed that the line of best fit was a straight line rather than a curve. Other students were penalised for 'double lines' or not extending the curve over the whole range of points.

83% of students were able to achieve at least two marks here.

- (c) A significant number of students did not draw a tangent at 0.95 g but used 0.95 g to calculate the mean rate over the first 0.95 g lost. Such students were able to score the third and fourth marking points. A small number of students drew the tangent in the wrong place choosing the other number (50) quoted in the question.

Some students had difficulty with the scales on the graph either by:

- forgetting that the y-axis intervals were 0.1 and calculating the difference as a whole number rather than the actual difference e.g. 18 rather than 0.18
- counting the number of squares on the y-axis or the x-axis e.g. $1.00 - 0.8 = 10$ (squares) rather than 0.2

Many students were able to state that the gradient of the tangent is calculated by the y value divided by the x value and calculate a value correctly. But some students neglected to quote their answer to two significant figures. 25% of students achieved full marks.